
3.5.6 The sequence of bases in DNA determines the structure of proteins, including enzymes.

The genetic code	The genetic code as base triplets in mRNA which code for specific amino acids. The genetic code is universal, non-overlapping and degenerate. The structure of molecules of messenger RNA (mRNA) and transfer RNA (tRNA). Candidates should be able to compare the structure and composition of DNA, mRNA and tRNA.
Polypeptide synthesis	Transcription as the production of mRNA from DNA. The role of RNA polymerase. The splicing of pre-mRNA to form mRNA in eukaryotic cells. Translation as the production of polypeptides from the sequence of codons carried by mRNA. The role of ribosomes and tRNA. Candidates should be able to • show understanding of how the base sequences of nucleic acids relate to the amino acid sequence of polypeptides, when provided with suitable data • interpret data from experimental work investigating the role of nucleic acids. Recall of specific codons and the amino acids for which they code, and of specific experiments, will not be tested.
Gene mutation	Gene mutations might arise during DNA replication. The deletion and substitution of bases. Gene mutations occur spontaneously. The mutation rate is increased by mutagenic agents. Some mutations result in a different amino acid sequence in the encoded polypeptide. Due to the degenerate nature of the genetic code, not all mutations result in a change to the amino acid sequence of the encoded polypeptide. The rate of cell division is controlled by proto-oncogenes that stimulate cell division and tumour suppressor genes that slow cell division. A mutated proto-oncogene, called an oncogene, stimulates cells to divide too quickly. A mutated tumour suppressor gene is inactivated, allowing the rate of cell division to increase.
